

BilenkinDSC670_Week 11_Milestone_4_Model_Update

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1 DSC670 – Milestone 4

1.1 Fine-Tuned GPT Model for Investment Research

This project develops a fine-tuned GPT-3.5 model designed to generate structured investment analysis based on SEC 10-K (annual report) style prompts.

The objective is to produce analyst-style summaries, risk assessments, and strategic insights to support investment decision-making.

The final deliverable includes an interactive Streamlit application that allows users to submit financial prompts and receive real-time responses from the fine-tuned model.

1.2 Project Overview

The objective of this project is to develop a fine-tuned GPT model capable of generating structured investment analysis from SEC 10-K filing prompts.

Large language models can generate general financial explanations; however, fine-tuning improves response consistency, structure, and domain-specific reasoning. By training the model on curated investment-style prompts and responses, the system is optimized to produce:

- Executive-style company summaries
- Identification of key financial and operational risks
- Strategic comparisons between competitors
- Clear, structured analyst-style outputs

This approach enhances reliability and ensures responses align with professional investment research standards.

1.3 Data Preparation

The training dataset was structured in JSONL format using prompt-response pairs compatible with OpenAI fine-tuning requirements.

Each training example follows a chat-based structure and includes:

- A **system message** defining the assistant as a professional investment analyst

- A **user message** containing a financial research prompt (e.g., 10-K summary, risk analysis, strategic comparison)
- An **assistant message** providing a structured analyst-style response

The dataset was curated to ensure clarity, consistency, and professional tone. Responses were designed to emphasize financial performance, risk factors, strategic positioning, and long-term investment considerations.

1.4 Training Dataset Creation

The following code constructs the structured training dataset in JSONL format. Each example follows the OpenAI chat fine-tuning schema and includes a system message, user prompt, and assistant response.

The dataset includes examples covering: - Executive summaries - Risk factor analysis - Strategic priorities - Financial performance evaluation - Competitive positioning - Regulatory and liquidity risks

The file is saved locally as `investment_training_data.jsonl` for use in the fine-tuning process.

```
[12]: import json

training_data = [
    {
        "messages": [
            {"role": "system", "content": "You are an investment analyst."},
            {"role": "user", "content": "Provide an executive summary of the
↪following 10-K excerpt: The company reported increased revenues driven by
↪strong demand in cloud services but faces rising operating costs."},
            {"role": "assistant", "content": "The company demonstrated solid
↪revenue growth primarily driven by expansion in its cloud services segment.
↪However, increasing operating costs may pressure margins and should be
↪monitored closely by investors."}
        ]
    },
    {
        "messages": [
            {"role": "system", "content": "You are an investment analyst."},
            {"role": "user", "content": "Analyze the risk factors: The company
↪depends heavily on a limited number of suppliers."},
            {"role": "assistant", "content": "Supplier concentration risk
↪presents a material vulnerability. Disruptions or pricing pressure from key
↪suppliers could negatively impact production capacity and profitability."}
        ]
    },
    {
        "messages": [
            {"role": "system", "content": "You are an investment analyst."},
```

```

        {"role": "user", "content": "Summarize strategic priorities:␣
↪Management plans expansion into international markets and increased R&D␣
↪spending."},
        {"role": "assistant", "content": "Management is prioritizing␣
↪international expansion and increased investment in research and␣
↪development, signaling a long-term growth strategy focused on innovation and␣
↪global market penetration."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Evaluate financial performance:␣
↪Revenue increased 10% year-over-year, but net income declined due to higher␣
↪expenses."},
        {"role": "assistant", "content": "While top-line growth remains␣
↪strong at 10% year-over-year, declining net income suggests cost pressures␣
↪are affecting profitability. Investors should assess whether these expenses␣
↪are temporary or structural."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Assess competitive positioning: The␣
↪company operates in a highly competitive market with rapid technological␣
↪change."},
        {"role": "assistant", "content": "Operating in a highly competitive␣
↪and rapidly evolving industry increases strategic risk. Sustained investment␣
↪in innovation will be critical to maintaining market share."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Explain the revenue model: The company␣
↪generates revenue through subscriptions and advertising."},
        {"role": "assistant", "content": "The company benefits from␣
↪diversified revenue streams, including recurring subscription income and␣
↪advertising. Subscription revenue enhances predictability, while advertising␣
↪introduces cyclical exposure."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},

```

```

        {"role": "user", "content": "Analyze margin pressures: Rising labor↵
↵and logistics costs are impacting gross margins."},
        {"role": "assistant", "content": "Increasing labor and logistics↵
↵expenses are compressing gross margins. Sustained cost inflation could↵
↵reduce operating leverage unless offset by pricing power or efficiency↵
↵improvements."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Discuss regulatory risks: New data↵
↵privacy laws may increase compliance costs."},
        {"role": "assistant", "content": "Evolving data privacy regulations↵
↵may increase compliance costs and operational complexity. Non-compliance↵
↵could also expose the company to legal and reputational risks."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Provide growth outlook analysis:↵
↵Management forecasts double-digit revenue growth next year."},
        {"role": "assistant", "content": "Management's projection of↵
↵double-digit revenue growth suggests continued demand strength. Investors↵
↵should evaluate underlying drivers to determine sustainability."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Evaluate capital allocation: The↵
↵company announced a stock buyback program."},
        {"role": "assistant", "content": "A stock buyback program may↵
↵signal management confidence in intrinsic value and can enhance shareholder↵
↵returns, though it reduces liquidity available for reinvestment."}
    ]
},
{
    "messages": [
        {"role": "system", "content": "You are an investment analyst."},
        {"role": "user", "content": "Analyze liquidity position: The↵
↵company holds significant cash reserves but has rising debt levels."},
        {"role": "assistant", "content": "Strong cash reserves provide a↵
↵liquidity cushion; however, increasing debt levels may elevate financial↵
↵risk if not matched by sustainable earnings growth."}
    ]
}

```

```

    ]
  },
  {
    "messages": [
      {"role": "system", "content": "You are an investment analyst."},
      {"role": "user", "content": "Summarize management discussion:␣
↪Management highlighted supply chain improvements and operational efficiency␣
↪initiatives."},
      {"role": "assistant", "content": "Management's focus on supply␣
↪chain optimization and efficiency initiatives indicates efforts to protect␣
↪margins and enhance operational resilience."}
    ]
  }
]

with open("investment_training_data.jsonl", "w") as f:
    for entry in training_data:
        f.write(json.dumps(entry) + "\n")

print("Training file created successfully.")

```

Training file created successfully.

1.5 Upload Training File

The following code uploads the JSONL training dataset to OpenAI in preparation for fine-tuning. The returned file ID is used to initiate the fine-tuning job in the next step.

```

[13]: # Upload training file to OpenAI for fine-tuning
from openai import OpenAI
client = OpenAI()

file = client.files.create(
    file=open("investment_training_data.jsonl", "rb"),
    purpose="fine-tune"
)

print("File ID:", file.id)

```

File ID: file-MQZMY2NKc4cSGaapRkYo91

1.6 Model Fine-Tuning

The base model selected for this project was **gpt-3.5-turbo-0125**, a conversational model optimized for chat-based interactions.

Fine-tuning was performed using the custom financial training dataset to improve:

- Response structure and clarity
- Financial reasoning consistency
- Domain-specific language and tone
- Alignment with professional investment analysis standards

By fine-tuning the model on curated 10-K style prompts, the system becomes more specialized compared to a general-purpose language model.

1.7 Create Fine-Tuning Job

The following code initiates the fine-tuning process using the uploaded training file.

A fine-tuning job ID is generated and can be used to monitor training progress until completion.

```
[14]: # Create fine-tuning job
job = client.fine_tuning.jobs.create(
    training_file="file-6caMfVxB5JLz6AW462ugvi",
    model="gpt-3.5-turbo"
)

print("Job ID:", job.id)
```

Job ID: ftjob-McMHDEdiNGcojaQhse4cH5NW

1.8 Fine-Tuning Job Summary

The fine-tuning job for `gpt-3.5-turbo-0125` completed successfully with the custom investment prompts.

- Fine-Tuned Model ID: `ft:gpt-3.5-turbo-0125:personal::D8zPtSHE`
- Training dataset: 12 prompt-response examples covering summaries, risks, strategic analysis, financial evaluation, and regulatory considerations.
- Training was completed without errors.

This fine-tuned model is used for all downstream testing and deployment in the Streamlit application.

1.8.1 Test the Fine-Tuned Model

Before integrating the model into the Streamlit app, we test it using a few sample prompts to ensure it produces investment-focused summaries and risk analyses. This helps verify that the fine-tuning worked as expected.

```
[15]: from openai import OpenAI

# Initialize OpenAI client
client = OpenAI()
```

```

# Fine-tuned model ID
fine_tuned_model_id = "ft:gpt-3.5-turbo-0125:personal::D8zPtSHE"

# Sample prompts
prompts = [
    "Summarize this 10-K filing for a long-term investor.",
    "Identify the top 5 risks mentioned in this 10-K filing.",
    "Compare this company's strategy to its main competitor."
]

# Generate outputs
for i, prompt in enumerate(prompts, start=1):
    response = client.chat.completions.create(
        model=fine_tuned_model_id,
        messages=[
            {"role": "system", "content": "You are an investment analyst."},
            {"role": "user", "content": prompt}
        ]
    )
    print(f"--- Output {i} ---")
    print(response.choices[0].message.content)
    print("\n")

```

--- Output 1 ---

This 10-K filing indicates that the company has maintained strong financial performance, with revenue growing steadily and consistent profitability. Operating efficiency improvements have supported margin expansion. Investors should note the company's strategic expansion into new markets and ongoing investments in technology to drive future growth. However, increased competition and regulatory risks should also be considered.

--- Output 2 ---

Based on the disclosed information, the top 5 risks for investors to consider are: 1) regulatory uncertainty, 2) dependence on key personnel, 3) industry competition, 4) cybersecurity threats, and 5) supply chain vulnerabilities. Investors should closely monitor how these risks may impact the company's long-term performance.

--- Output 3 ---

This company is prioritizing product innovation and premium pricing to drive revenue growth. In contrast, its main competitor is focused on cost leadership and operational efficiency to maintain market share.

1.8.2 Build a Streamlit Application for Investment Research

This cell sets up a Streamlit interface to interact with the fine-tuned model. Users can input prompts related to SEC 10-K filings and receive analyst-style summaries, risk assessments, and strategic insights.

The application allows:

- Input of custom prompts about financial filings.
- Generation of responses using the fine-tuned investment research model.
- Display of analyst-style output directly in the browser.

Run this cell using `streamlit run app.py` in a terminal to open the interactive app in your browser.

```
[16]: import streamlit as st
from openai import OpenAI

# Initialize OpenAI client
client = OpenAI()

fine_tuned_model_id = "ft:gpt-3.5-turbo-0125:personal::D8zPtSHE"

st.title("Investment Research Assistant")
st.write("Enter prompts related to SEC 10-K filings and get analyst-style_
↳ summaries and risk assessments.")

# User input
user_prompt = st.text_area("Enter your prompt here:", "")

if st.button("Generate Response"):
    if user_prompt.strip() == "":
        st.warning("Please enter a prompt before generating a response.")
    else:
        with st.spinner("Generating response..."):
            response = client.chat.completions.create(
                model=fine_tuned_model_id,
                messages=[
                    {"role": "system", "content": "You are an investment_
↳ analyst."},
                    {"role": "user", "content": user_prompt}
                ]
            )
            output = response.choices[0].message.content
            st.subheader("Analyst Response")
            st.write(output)
```


1.9 Conclusion

The fine-tuned model successfully generates structured investment analysis responses. The Streamlit application provides an accessible interface for interacting with the model. Future improvements could include incorporating real SEC filing data and expanding the training dataset to enhance financial reasoning depth.